Solutions to the exercises in *Categorical Logic and Type Theory* by Bart Jacobs

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Updated: 2023-04-23T02:28:07Z

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Chapter 0

Prospectus

0.2 The logic and type theory of sets

0.2.1. Define a left adjoint $F : \mathrm{Fam}(\mathbf{Sets}) \to \mathbf{Pred}$ to the inclusion functor

$$\begin{tikzcd}
\mathbf{Pred} \arrow[rr, hook] \arrow[dr] & & \mathrm{Fam}(\mathbf{Sets}) \arrow[ll, bend right, "F"'] \arrow[dl] \\
& \mathbf{Sets} &
\end{tikzcd}$$

such that: (1) F makes the triangle commute (so it does not change the index set), and (2) F commutes with substitution.

Solution. Recall that the inclusion functor $G : \mathbf{Pred} \hookrightarrow \mathrm{Fam}(\mathbf{Sets})$ sends a predicate (I, X) (where $X \subseteq I$) to the family $G(I, X) = (X_i)_{i \in I}$, where

$$X_i = \begin{cases} \{*\} & \text{if } i \in X \\ \emptyset & \text{otherwise.} \end{cases}$$

Define the functor $F : \mathrm{Fam}(\mathbf{Sets}) \to \mathbf{Pred}$ on objects by

$$F\left((X_i)_{i \in I}\right) = \left(I, \left\{i \in I : X_i \neq \emptyset\right\}\right)$$

and on morphisms by

$$F\left((X_i)_{i \in I} \xrightarrow{(u,f)} (Y_j)_{j \in J}\right) = u.$$

Note that F is well-defined on morphisms. Indeed, suppose $(X_i)_{i \in I}$ and $(Y_j)_{j \in J}$ are families of sets and $(u, f) : (X_i)_{i \in I} \to (Y_j)_{j \in J}$ is a morphism of families. If $i \in I$ and $X_i \neq \emptyset$, then $\emptyset \neq f(X_i) \subseteq Y_{u(i)}$, so $Y_{u(i)} \neq \emptyset$, and hence

u is a morphism from $F\left((X_i)_{i\in I}\right)$ to $F\left((Y_j)_{j\in J}\right)$ in **Pred**. Moreover, F does not change the index set by definition.

Next, we claim that $F \dashv G$ by demonstrating the adjointness criteria in [2, Definition 3.1.4], namely that there is a "unit" natural transformation

$$\eta : \mathrm{id}_{\mathrm{Fam}(\mathbf{Sets})} \longrightarrow G \circ F.$$

such that for each I-indexed family of sets $(X_i)_{i\in I}$, each object (J, Y) in **Pred**, and each morphism $(v, g) : (X_i)_{i\in I} \to G(J, Y)$ in Fam(**Sets**), there exists a unique morphism $v' : F\left((X_i)_{i\in I}\right) \to (J, Y)$ in **Pred** such that $G(v') \circ \eta_{(X_i)_{i\in I}} = (v, g)$, as in the following commutative diagram.

$$\begin{tikzcd}
\textbf{Pred} & \mathrm{Fam}(\textbf{Sets}) \\
F\left((X_i)_{i\in I}\right) \arrow[d, dashed, "v'"] & (X_i)_{i\in I} \arrow[r, "\eta_{(X_i)_{i\in I}}"] \arrow[dr, "(v,g)"'] & G\left(F\left((X_i)_{i\in I}\right)\right) \arrow[d, dashed, "G(v')"] \\
(J, Y) & & G(J, Y)
\end{tikzcd}$$

To this end, for an object $(X_i)_{i\in I}$ define the component

$$\eta_{(X_i)_{i\in I}} : (X_i)_{i\in I} \longrightarrow G\left(F\left((X_i)_{i\in I}\right)\right) = \left(\text{if } X_i \neq \emptyset \text{ then } \{*\} \text{ else } \emptyset\right)_{i\in I}$$

as

$$\eta_{(X_i)_{i\in I}} = \left(\mathrm{id}_I, \left(f_i : X_i \longrightarrow \text{if } X_i \neq \emptyset \text{ then } \{*\} \text{ else } \emptyset\right)_{i\in I}\right),$$

where

$$f_i = \begin{cases} x \mapsto * & \text{if } X_i \neq \emptyset \\ \mathrm{id}_\emptyset & \text{otherwise.} \end{cases}$$

We omit the rest of the verification that η is the unit of the adjunction $F \dashv G$.

Finally, we claim that F commutes with substitution. Indeed, if $u : I \to J$ is a morphism in **Sets** and $(Y_i)_{j\in J}$ is a J-indexed family of sets, then

$$\begin{aligned}
u^*\left(F\left((Y_j)_{j\in J}\right)\right) &= u^*\left(J, \left\{j \in J : Y_j \neq \emptyset\right\}\right) \\
&= \left(I, \left\{i \in I : Y_{u(i)} \neq \emptyset\right\}\right) \\
&= F\left(\left(Y_{u(i)}\right)_{i\in I}\right) \\
&= F\left(u^*\left((Y_j)_{j\in J}\right)\right).
\end{aligned}$$

□

0.2.2. Define for a subset $X \subseteq I$ the relation $\mathrm{nEq}(X) \subseteq I \times I$ by

$$\mathrm{nEq}(X) = \Big\{(i, i') : i \neq i' \text{ or } i \in X\Big\}$$

and show that the assignment $X \mapsto \mathrm{nEq}(X)$ is right adjoint to contraction $\delta^* : P(I \times I) \to P(I)$. Notice that $\mathrm{nEq}(X)$ at the bottom element $X = \emptyset$ is inequality on I.

Solution. Fix a set I and consider the functors

$$P(I) \underset{\mathrm{nEq}}{\overset{\delta^*}{\leftrightarrows}} P(I \times I).$$

To show that $\delta^* \dashv \mathrm{nEq}$, we fix $X \subseteq I$ and $Y \subseteq I \times I$ and claim that $\delta^*(Y) \subseteq X$ if and only if $Y \subseteq \mathrm{nEq}(X)$. Suppose $\delta^*(Y) \subseteq X$ and $(i, j) \in Y$. If $i \neq j$, then $(i, j) \in \mathrm{nEq}(X)$, and if $i = j$, then $i \in \delta^*(Y) \subseteq X$, and hence $(i, j) \in \mathrm{nEq}(X)$. Thus, $Y \subseteq \mathrm{nEq}(X)$. Conversely, suppose $Y \subseteq \mathrm{nEq}(X)$ and $i \in \delta^*(Y)$. Then $(i, i) \in Y \subseteq \mathrm{nEq}(X)$, so $i \in X$, whence $\delta^*(Y) \subseteq X$ and $(i, j) \in Y$. $\square$

0.2.3. Show that the equality functor $\mathrm{Eq} : \mathbf{Sets} \to \mathbf{Rel}$ also has a right adjoint.

Solution. Define the functor

$$F : \mathbf{Rel} \longrightarrow \mathbf{Sets}$$

on objects by

$$F(I, R) = \{i \in I : (i, i) \in R\}$$

and on morphisms by

$$F\Big((I, R) \xrightarrow{u} (J, S)\Big) = u|_{F(I,R)}.$$

Note that F is well-defined on morphisms since if $u : (I, R) \to (J, S)$ is a morphism in $\mathbf{Rel}$ and $i \in F(I, R)$, then $(i, i) \in R$, so $(u(i), u(i)) \in S$, and hence $u(i) \in F(J, S)$. We claim that $\mathrm{Eq} \dashv F$. Fix an object I in $\mathbf{Sets}$ and an object (J, S) in $\mathbf{Rel}$. Define functions

$$f : \mathbf{Rel}(\mathrm{Eq}(I), (J, S)) \longrightarrow \mathbf{Sets}(I, F(J, S))$$
$$g : \mathbf{Sets}(I, F(J, S)) \longrightarrow \mathbf{Rel}(\mathrm{Eq}(I), (J, S))$$

by

$$f\left(\mathrm{Eq}(I) \xrightarrow{u} (J,S)\right) = u \qquad\qquad g\left(I \xrightarrow{u} F(J,S)\right) = u.$$

Note that f is well-defined since if $u : \mathrm{Eq}(I) \to (J,S)$ is a morphism in **Rel** and $i \in I$, then $(i,i) \in \mathrm{Eq}(I)$, so $(u(i), u(i)) \in S$, and hence $u(i) \in F(J,S)$. Similarly, note that g is well-defined since if $u : I \to F(J,S)$ is a morphism in **Sets** and $i \in I$, then $u(i) \in F(J,S)$, so $(u(i), u(i)) \in S$. Thus, f and g are mutual inverses, so we have

$$\mathbf{Rel}(\mathrm{Eq}(I), (J,S)) \cong \mathbf{Sets}(I, F(J,S)).$$

We omit the proof of naturality of this bijection and conclude that $\mathrm{Eq} \dashv F$. $\square$

0.2.4. Check that the operation $(I,X) \mapsto \{I \mid X\}$ yields a functor $\mathrm{Fam}(\mathbf{Sets}) \to \mathbf{Sets}$, and show that it is right adjoint to the terminal object functor $\mathbf{Sets} \to \mathrm{Fam}(\mathbf{Sets})$, mapping a set J to the family of singletons $(\{*\})_{j \in J}$. Describe the unit and counit of the adjunction explicitly.

Solution. Define the functor $R : \mathrm{Fam}(\mathbf{Sets}) \to \mathbf{Sets}$ on objects by

$$R(I,X) = \{I \mid X\} = \sum_{i \in I} X_i = \Big\{(i,x) : i \in I \text{ and } x \in X_i\Big\}$$

and on morphisms by

$$R\left((I,X) \xrightarrow{(u,f)} (J,Y)\right) = \Big((i,x) \longmapsto (u(i), f_i(x))\Big).$$

We claim that $L \dashv R$, where $L : \mathbf{Sets} \to \mathrm{Fam}(\mathbf{Sets})$ is the "terminal object functor," defined on objects by $L(J) = (\{*\})_{j \in J}$. It suffices to prove that there is an isomorphism

$$\mathrm{Fam}(\mathbf{Sets})\left((\{*\})_{j \in J}, (I,X)\right) \cong \mathbf{Sets}\left(J, \{I \mid X\}\right),$$

which is natural in J in **Sets** and (I,X) in $\mathrm{Fam}(\mathbf{Sets})$. Indeed, define

$$\varphi : \mathrm{Fam}(\mathbf{Sets})\left((\{*\})_{j \in J}, (I,X)\right) \longrightarrow \mathbf{Sets}\left(J, \{I \mid X\}\right)$$

by

$$\varphi\left(\left(\{*\}\right)_{j\in J} \xrightarrow{(u,f)} (I,X)\right) = \Big(j \longmapsto (u(j), f_j(*))\Big).$$

One can verify that φ is a bijection, with

$$\varphi^{-1}\left(J \xrightarrow{g} \{I \mid X\}\right) = \left(j \longmapsto \mathrm{fst}(g(j)), \Big(* \longmapsto \mathrm{snd}(g(j))\Big)_{j\in J}\right),$$

and that φ is natural in J and (I,X). Thus, it yields an adjunction $L \dashv R$. The unit of this adjunction is then given by

$$\eta_J = \varphi\left(\mathrm{id}_{(\{*\})_{j\in J}}\right) : J \longrightarrow \sum_{j\in J}\{*\}, \qquad \eta_J(j) = (j,*),$$

and the counit is given by

$$\begin{aligned}\varepsilon_{(I,X)} &= \varphi^{-1}\left(\mathrm{id}_{\{I|X\}}\right) : \left(\{*\}\right)_{(i,x)\in\{I|X\}} \longrightarrow (I,X)\\ \varepsilon_{(I,X)} &= \left((i,x) \longmapsto i, \Big(* \longmapsto x\Big)_{(i,x)\in\{I|X\}}\right).\end{aligned}$$

□

Chapter 1

Introduction to fibred category theory

1.1 Fibrations

1.1.1.

(i) Prove Proposition 1.1.4.[a]

(ii) Suppose f is Cartesian and g and h are above the same map. Show that $f \circ g = f \circ h$ implies $g = h$.

[a] [6, Proposition 1.1.4] asserts that Cartesian liftings are unique up-to-isomorphism (in a slice): if f and f' with $\operatorname{cod} f = \operatorname{cod} f'$ are both Cartesian over the same map, then there is a unique vertical isomorphism $\varphi : X \xrightarrow{\cong} X'$ with $f' \circ \varphi = f$.

Remark 1.1.1.1. We will prove this in the more general context of displayed categories (cf. [1]). The conclusion of Exercise 1.1.1(i) (i.e., [6, Proposition 1.1.4]) appears in the literature in, e.g., [3, Lemma 8.1.4(2)], [4, Remark 3.3], [5, Lemma 2.3(ii)], and [7, Proposition 9.1.4(4)].

Solution, (i). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ is a morphism in $\mathbb{B}$, and $f : X \to Y$ and $f' : X' \to Y$ are Cartesian morphisms in $\mathbb{E}$ over u. Since f' is Cartesian and f is a morphism into Y over $u = u \circ \mathrm{id}_I$, there exists a unique morphism $\varphi : X \to X'$ in $\mathbb{E}$ over I such that $f = f' \circ \varphi$. This is summarized by the following diagram.

$$
\begin{tikzcd}
X \arrow[d] \arrow[rr, bend left, "f"] & & \\
I \arrow[rr, bend left] \arrow[dr, equal] & X' \arrow[r, "f'"] \arrow[d] \arrow[dr, bend left, "u"] & Y \arrow[d] \\
& I \arrow[r, "u"] & J
\end{tikzcd}
\quad \Longrightarrow \quad
\begin{tikzcd}
X \arrow[d] \arrow[rr, bend left, "f"] \arrow[dr, dashed, "\exists!\,\varphi"] & & \\
I \arrow[dr, equal] & X' \arrow[r, "f'"] \arrow[d] & Y \arrow[d] \\
& I \arrow[r, "u"] & J
\end{tikzcd}
$$

Similarly, there exists a unique morphism $\psi : X' \to X$ in $\mathbb{E}$ over I such that $f' = f \circ \psi$. Moreover, both $\mathrm{id}_{X'}$ and $\varphi \circ \psi$ are morphisms in $\mathbb{E}$ over I whose composition with f' is f' itself, as suggested in the following diagram.

$$\begin{tikzcd}
X' \arrow[dr, "\psi"] \arrow[drr, bend left, "f'"] \arrow[ddr, bend right, "\mathrm{id}_{X'}"'] \arrow[ddd] & & & \\
& X \arrow[dr, "\varphi"] \arrow[drr, bend left, "f"] \arrow[dd] & & \\
I \arrow[dr, equal] & & X' \arrow[r, "f'"] \arrow[dd] & Y \arrow[dd] \\
& I \arrow[dr, equal] & & \\
& & I \arrow[r, "u"] & J
\end{tikzcd}$$

Thus, since f' is Cartesian, $\varphi \circ \psi = \mathrm{id}_{X'}$. By symmetry, $\psi \circ \varphi = \mathrm{id}_X$, and hence φ is an isomorphism. $\square$

Solution, (ii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ and $v : K \to I$ are morphisms in $\mathbb{B}$, $f : X \to Y$ is a Cartesian morphism in $\mathbb{E}$ over u, and $g, h : Z \to X$ are morphisms in $\mathbb{E}$ over v such that $f \circ g = f \circ h$. These suppositions are summarized in the following commutative diagram.

$$\begin{tikzcd}
Z \arrow[dr, bend left, "g"] \arrow[drr, bend left, "f\circ g = f\circ h"] \arrow[dr, bend right, "h"'] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"] \arrow[drr, bend left, "u\circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Since f is Cartesian, this situation immediately implies that $g = h$. $\square$

1.1.2. Let $p : \mathbb{E} \to \mathbb{B}$ be a functor; assume $f : X \to Y$ is in $\mathbb{E}$ and put $u = pf$. Show that f is Cartesian if and only if for each $Z \in \mathbb{E}$ and $v : pZ \to pX$ in $\mathbb{B}$, the function

$$\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$$

is an isomorphism.

Remark 1.1.2.1. We will prove this in the more general context of displayed categories (cf. [1]). However, this result is essentially an unpacking of the definition of Cartesianness and, in particular, of [1, Definition 5.1].

Solution. Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ is a morphism in $\mathbb{B}$, and $f : X \to Y$ is a morphism in $\mathbb{E}$ over u.

($\Longrightarrow$) Suppose f is Cartesian, and take a morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K. Since f is Cartesian, this situation gives rise to a one-to-one correspondence between the dashed arrows in the diagram below.

$$\begin{tikzcd}
Z \arrow[dd] \arrow[dr, dashed] \arrow[drr, dashed, bend left] & & \\
& X \arrow[r, "f"] \arrow[dd] \arrow[ddr, bend left, "u\circ v"] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[urr, bend left] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Indeed, given a morphism $g : Z \to X$ in $\mathbb{E}$ over v, composition with f yields a morphism $f \circ g : Z \to Y$ over $u \circ v$. Conversely, given a morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$, there is a unique morphism $g : Z \to X$ in $\mathbb{E}$ over v such that $f \circ g = h$ since f is Cartesian. This is precisely the assertion that $\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$ is an isomorphism (of sets).

($\Longleftarrow$) Conversely, suppose that $\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$ is an isomorphism for every morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K. Consider a particular morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K, and suppose there is a morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. Then the assumed isomorphism implies that there is a unique morphism $g : Z \to X$ in $\mathbb{E}$ over v such that $h = f \circ g$, which shows that f is Cartesian. $\square$

1.1.3. Consider the total category of a fibration. Show that

(i) every morphism factors as a vertical map followed (diagrammatically) by a Cartesian one;

(ii) a Cartesian map above an isomorphism is an isomorphism. Espe-

cially a vertical Cartesian map is an isomorphism.

Solution, (i). Let $\mathbb{E}$ be a fibration over $\mathbb{B}$, and let $f : X \to Y$ be a morphism in $\mathbb{E}$ over a morphism $u : I \to J$ in $\mathbb{B}$. Then there exists a Cartesian morphism $f' : X' \to Y$ in $\mathbb{E}$ above u. Then we are in the situation of the diagram on the left below, so there exists a unique (vertical) morphism $g : X \to X'$ over I such that the diagram below on the right commutes.

$$\begin{tikzcd}
X \arrow[dr, bend left, "f"] \arrow[dd] & & \\
& X' \arrow[r, "f'"] \arrow[dd] \arrow[dr, bend left, "u"] & Y \arrow[dd] \\
I \arrow[dr, equal] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}
\Longrightarrow
\begin{tikzcd}
X \arrow[dr, bend left, "f"] \arrow[dr, dashed, "g"] \arrow[dd] & & \\
& X' \arrow[r, "f'"] \arrow[dd] \arrow[dr, bend left, "u"] & Y \arrow[dd] \\
I \arrow[dr, equal] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, $f = f' \circ g$; in other words, f factors as a vertical morphism followed by a Cartesian morphism. $\square$

Solution, (ii). Let $\mathbb{E}$ be a fibration over $\mathbb{B}$, and let $f : X \to Y$ be a Cartesian morphism in $\mathbb{E}$ over an isomorphism $u : I \to J$ in $\mathbb{B}$. Let $v = u^{-1} : J \to I$ in $\mathbb{B}$, and consider the situation in the following commutative diagram.

$$\begin{tikzcd}
Y \arrow[dr, dashed, "\exists! g"] \arrow[drr, bend left, "\mathrm{id}_Y"] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] \arrow[dr, bend left, "\mathrm{id}_J"] & Y \arrow[dd] \\
J \arrow[dr, "\sim", "v"'] & & \\
& I \arrow[r, "\sim", "u"'] & J
\end{tikzcd}$$

Here the dashed morphism $g : Y \to X$ exists uniquely by virtue of f being Cartesian. Since $f \circ g = \mathrm{id}_Y$ is Cartesian (by Exercise 1.1.4(i)) and f is Cartesian, Exercise 1.1.4(iii) implies that g is Cartesian over the isomorphism v, and $f \circ g = \mathrm{id}_Y$. The same argument then implies that there exists a morphism $h : X \to Y$ over $u = v^{-1}$ such that $g \circ h = \mathrm{id}_X$. But then

$$f = f \circ \mathrm{id}_X = f \circ g \circ h = \mathrm{id}_Y \circ h = h,$$

so $g \circ f = \mathrm{id}_X$, and hence $g = f^{-1}$, so f is an isomorphism. $\square$

1.1.4. Let $\begin{array}{c}\mathbb{E}\\ \downarrow p\\ \mathbb{B}\end{array}$ be a fibration. Prove that

(i) all isomorphisms in $\mathbb{E}$ are Cartesian;

(ii) if $\xrightarrow{f}\xrightarrow{g}$ is a composable pair of Cartesian morphisms in $\mathbb{E}$, then also their composite $\xrightarrow{g\circ f}$ is Cartesian.

Hence it makes sense to talk about the subcategory $\mathrm{Cart}(\mathbb{E}) \hookrightarrow \mathbb{E}$ having all objects from $\mathbb{E}$ but only the Cartesian arrows. We write $|p|$ for the composite $\mathrm{Cart}(\mathbb{E}) \hookrightarrow \mathbb{E} \to \mathbb{B}$.

(iii) Let $\xrightarrow{f}\xrightarrow{g}$ be a composable pair in $\mathbb{E}$ again. Show now that if g and $g \circ f$ are Cartesian, then f is Cartesian as well.

(iv) Verify that a consequence of (iii) is that the functor $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$ is a fibration. From (ii) in the previous exercise it follows that all fibres of $|p|$ are groupoids (*i.e.* that all maps in the fibres are isomorphisms).

[This $|p|$ will be called the **fibration of objects** of p.]

Remark 1.1.4.1. We will prove this in the more general context of displayed categories (cf. [1]).

Solution, (i). Let $\mathbb{E}$ be a displayed category over $\mathbb{B}$ (not necessarily a fibration), and let $f : X \to Y$ be a displayed isomorphism in $\mathbb{E}$ above an isomorphism $u : I \to J$ in $\mathbb{B}$. Suppose $v : K \to I$ is a morphism in $\mathbb{B}$ and $g : Z \to Y$ is a morphism in $\mathbb{E}$ above $u \circ v : K \to J$. Then $f^{-1} \circ g : Z \to X$ is the unique morphism in $\mathbb{E}$ above v such that the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "f^{-1}\circ g"] \arrow[drr, bend left, "g"] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend left, "u\circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Indeed, to verify uniqueness, suppose $h : Z \to X$ is another morphism in $\mathbb{E}$ above v such that $f \circ h = g$, and observe that

$$h = f^{-1} \circ f \circ h = f^{-1} \circ g.$$

Therefore f is Cartesian. $\square$

Remark 1.1.4.2. The conclusion of Exercise 1.1.4(ii) appears in the literature in, e.g., [3, Lemma 8.1.4(i)]

Solution, (ii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$ (not necessarily a fibration), $u : I \to J$ and $v : J \to K$ are morphisms in $\mathbb{B}$, and $f : X \to Y$ and $g : Y \to Z$ are Cartesian morphisms in $\mathbb{E}$ over u and v, respectively. To show that $g \circ f : X \to Z$ is Cartesian over $v \circ u : I \to K$, consider a morphism $w : L \to I$ in $\mathbb{B}$ and a morphism $h : W \to Z$ in $\mathbb{E}$ above $v \circ u \circ w$. These data are summarized in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[rrr, bend left, "h"] \arrow[d] & & & \\
 & X \arrow[r, "f"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[r, "g"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Z \arrow[d] \\
L \arrow[r, "w"] & I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

Since g is Cartesian, there exists a unique morphism $h' : W \to Y$ in $\mathbb{E}$ above $u \circ w$ such that $h = g \circ h'$, as in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[rrr, bend left, "h"] \arrow[rr, bend left, dashed, "h'"] \arrow[d] & & & \\
 & X \arrow[r, "f"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[r, "g"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Z \arrow[d] \\
L \arrow[r, "w"] & I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

Moreover, since f is Cartesian, there exists a unique morphism $h'' : W \to X$

in $\mathbb{E}$ above w such that $h' = f \circ h''$, as in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[dr, dashed, "h''"] \arrow[drr, bend left, "h'"] \arrow[drrr, bend left, "h"] \arrow[dd] & & & \\
& X \arrow[r, "f"] \arrow[dd] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[r, "g"] \arrow[dd] \arrow[dr, phantom, "\lrcorner"] & Z \arrow[dd] \\
L \arrow[dr, "w"] & & & \\
& I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

In particular, we have

$$h = g \circ h' = g \circ f \circ h''.$$

It remains to show that if $i : W \to X$ is another morphism in $\mathbb{E}$ above w such that $h = g \circ f \circ i$, then $i = h''$. Indeed, in this case the uniqueness of h' implies that $f \circ i = h'$, and so the uniqueness of h'' implies that $i = h''$. Thus, $g \circ f$ is Cartesian. $\square$

Solution, (iii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$ (not necessarily a fibration), $u : I \to J$ and $v : J \to K$ are morphisms in $\mathbb{B}$, and $f : X \to Y$ and $g : Y \to Z$ are morphisms in $\mathbb{E}$ over u and v, respectively, such that g and $g \circ f$ are Cartesian. To show that $f : X \to Y$ is Cartesian over u, consider a morphism $w : L \to I$ in $\mathbb{B}$ and a morphism $h : W \to Y$ in $\mathbb{E}$ above $u \circ w$. This situation is summarized in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[drrr, bend left, "h"] \arrow[dd] & & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[r, "g"] \arrow[dd] & Z \arrow[dd] \\
L \arrow[dr, "w"] & & & \\
& I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

But then $g \circ h : W \to Z$ is a morphism in $\mathbb{E}$ above $v \circ u \circ w$, so since $g \circ f$ is Cartesian, there exists a unique morphism $h' : W \to X$ in $\mathbb{E}$ over w such that $g \circ h = g \circ f \circ h'$. Since g is Cartesian, Exercise 1.1.1(i) implies that $h = f \circ h'$. Moreover, suppose $i : W \to X$ is another morphism above w such that $h = f \circ i$. Then $g \circ h = g \circ f \circ i$, so the uniqueness of h' arising from $g \circ f$ being Cartesian implies that $i = h'$. Thus, f is Cartesian. $\square$

Solution, (iv). Let $p : \mathbb{E} \to \mathbb{B}$ be a fibration; we will show that $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$ is also a fibration. Consider a morphism $u : I \to J$ in $\mathbb{B}$ and an object Y in $\mathbb{E}$ over J. Since p is a fibration, there exists a p-Cartesian morphism $f : X \to Y$ in $\mathbb{E}$ over u. We claim that f is $|p|$-Cartesian as well. Indeed, suppose there is a morphism $v : K \to I$ in $\mathbb{B}$ and a p-Cartesian morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. Since f is p-Cartesian, there exists a unique morphism $h : Z \to X$ in $\mathbb{E}$ over v such that $g = f \circ h$. But now (iii) implies that h is p-Cartesian as well, so the entire diagram below is actually a diagram in the displayed category induced by $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "\exists! h"] \arrow[drr, bend left, "g"] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend left, "u \circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Therefore, f is $|p|$-Cartesian, so $|p|$ is a fibration. $\square$

1.1.5. Verify that the following two results—known as the Pullback Lemmas—are a consequence of (ii) and (iii) in the previous exercise. Consider

$$\begin{tikzcd}
\bullet \arrow[r] \arrow[d] \arrow[dr, phantom, "(B)"] & \bullet \arrow[r] \arrow[d] \arrow[dr, phantom, "(A)"] & \bullet \arrow[d] \\
\bullet \arrow[r] & \bullet \arrow[r] & \bullet
\end{tikzcd}$$

(i) If (A) and (B) are pullback squares, then the outer rectangle is also a pullback square.

(ii) If the outer rectangle and (A) are pullback squares, then (B) is a pullback square as well.

Solution. Recall (e.g., from [4, Example 3.15] or [6, Proposition 1.1.6(ii)])

that in a category $\mathbb{B}$, a commutative square

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "p"] & Y \arrow[d, "q"] \\ I \arrow[r, "u"] & J \end{tikzcd}$$

is a pullback square if and only if (u, f) is a Cartesian lift of u in the context of the codomain functor $\mathrm{cod} : \mathbb{B}^{\to} \to \mathbb{B}$, as in the following diagram.

$$\begin{tikzcd} \begin{pmatrix} X \\ \downarrow p \\ I \end{pmatrix} \arrow[r, "(u,f)"] \arrow[d, "\mathrm{cod}"] & \begin{pmatrix} Y \\ \downarrow q \\ J \end{pmatrix} \arrow[d, "\mathrm{cod}"] \\ I \arrow[r, "u"] & J \end{tikzcd}$$

Therefore, (i) and (ii) follow immediately from Exercise 1.1.4(ii) and Exercise 1.1.4(iii), respectively. □

1.1.6. Consider a functor $p : \mathbb{E} \to \mathbb{B}$. We describe a slightly weaker notion of Cartesianness, than the one above. Call a morphism $f : X \to Y$ in $\mathbb{E}$ **weak Cartesian** if for each $g : Z \to Y$ with $pf = pg$ there is a unique vertical $h : Z \to X$ with $f \circ h = g$. Show that the functor p is a fibration if and only if both

(a) every morphism $u : I \to pY$ in $\mathbb{B}$ has a weak Cartesian lifting $f : X \to Y$;

(b) the composition of two weak Cartesian morphisms is again weak Cartesian.

Remark 1.1.6.1. The conclusion of Exercise 1.1.6 appears in the literature in, e.g., [3, Proposition 8.1.7] and [5, Proposition 2.5].

Solution. ($\Longrightarrow$) Since every Cartesian morphism is weak Cartesian and compositions of Cartesian morphisms are Cartesian by Exercise 1.1.4(ii), every fibration satisfies (a) and (b).

($\Longrightarrow$) Conversely, suppose (a) and (b) hold. Let $u : I \to J$ be a morphism in $\mathbb{B}$, and let Y be an object in $\mathbb{E}$ above J. By (a), let $f : X \to Y$ be a weak Cartesian lift of u in $\mathbb{E}$. Consider another morphism $v : K \to I$ in $\mathbb{B}$ and a morphism $g : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. By (a), choose a weak Cartesian lift $p : W \to X$ in $\mathbb{E}$ of v. Now $f \circ p$ is weak Cartesian over $u \circ v$ by (b), so there exists a unique vertical morphism $q : Z \to W$ over K such that the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "q"] \arrow[drrr, bend left, "g"] \arrow[dd] & & & \\
& W \arrow[r, "p"] \arrow[dd] & X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, equal] & & & \\
& K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, "p\circ q"] \arrow[drrr, bend left, "g"] \arrow[dd] & & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"] & & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

It remains to show that if $h : Z \to X$ is another morphism in $\mathbb{E}$ above v such that $g = f \circ h$, then $h = p \circ q$. In this case, since p is weak Cartesian, we can find a unique vertical morphism $q' : Z \to W$ over K such that $h = p \circ q'$.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "q'"] \arrow[drr, bend left, "h"] \arrow[drrr, bend left, "g"] \arrow[dd] & & & \\
& W \arrow[r, "p"] \arrow[dd] & X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, equal] & & & \\
& K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, $g = f \circ p \circ q'$, so the uniqueness of q implies that $q' = q$, and hence $h = p \circ q$. Thus, f is a Cartesian lift of u, so p is a fibration. $\square$

1.1.7. Check that the following are (trivial) examples of fibrations

$$\begin{tikzcd} \mathbb{B} \times \mathbb{C} \arrow[d, "\mathrm{fst}"] \\ \mathbb{B} \end{tikzcd} \qquad \begin{tikzcd} \mathbb{B} \arrow[d, "\mathrm{id}"] \\ \mathbb{B} \end{tikzcd} \qquad \begin{tikzcd} \mathbb{B} \arrow[d] \\ 1 = \{*\} \end{tikzcd} \qquad \begin{tikzcd} X \arrow[d] \\ I \end{tikzcd}$$

where X, I are sets (*i.e.* discrete categories).

Solution. Consider the functor $\mathbb{B} \times \mathbb{C} \xrightarrow{\mathrm{fst}} \mathbb{B}$, which projects onto the first component. Consider a morphism $u : I \to J$ in $\mathbb{B}$ and an object (J, Y) in $\mathbb{B} \times \mathbb{C}$ above J. We claim that $(v, \mathrm{id}_Y) : (I, Y) \to (J, Y)$ is a Cartesian lift of u. Suppose there are morphisms $v : K \to J$ in $\mathbb{B}$ and $(w, g) : (K, Z) \to (J, Y)$ in $\mathbb{B} \times \mathbb{C}$ above $u \circ v$ (i.e., $w = u \circ v$), as in the following diagram.

$$\begin{tikzcd} (K,Z) \arrow[dr, dashed, "?"] \arrow[drr, bend left, "(w,g)"] \arrow[dd, "\mathrm{fst}"] & & \\ & (I,Y) \arrow[r, "(u,\mathrm{id}_Y)"] \arrow[d, "\mathrm{fst}"] & (J,Y) \arrow[d, "\mathrm{fst}"] \\ K \arrow[r, "v"] & I \arrow[r, "u"] & J \end{tikzcd}$$

Observe that $(v, g) : (K, Z) \to (I, Y)$ is the unique morphism in $\mathbb{B} \times \mathbb{C}$ that fills in the dashed "?" morphism making the diagram above commute, so (u, id_Y) is Cartesian and hence $\mathbb{B} \times \mathbb{C} \xrightarrow{\mathrm{fst}} \mathbb{B}$ is a fibration.

Next, consider the identity functor $\mathbb{B} \xrightarrow{\mathrm{id}} \mathbb{B}$, and observe that this situation is equivalent to the functor $\mathbb{B} \times 1 \xrightarrow{\mathrm{fst}} \mathbb{B}$, so it is a fibration by the first part of this exercise. More explicitly, consider a morphism $u : I \to J$ in $\mathbb{B}$. The only object above J in this scenario is J itself, and we claim that u itself is the unique Carteisan lift of u. Indeed, given morphisms $v : K \to I$ in $\mathbb{B}$ ("below") and $w : K \to J$ in $\mathbb{B}$ ("above") over $u \circ v$ (i.e., $w = u \circ v$), the

unique morphism $K \to I$ which fills in the dashed "?" arrow in the diagram below is v itself.

$$\begin{tikzcd} K \arrow[rr, bend left, "w"] \arrow[dr, dashed, bend left, "?"] \arrow[dr, bend right, "v"'] & & J \\ & I \arrow[ur, "u"] & \end{tikzcd}$$

Thus, u is a Cartesian lift of u, and $\mathbb{B} \xrightarrow{\mathrm{id}} \mathbb{B}$ is a fibration.

Consider the functor $\mathbb{B} \to 1$, and observe that this situation is equivalent to the functor $1 \times \mathbb{B} \xrightarrow{\mathrm{fst}} 1$, so it is a fibration by the first part of this exercise. More explicitly, consider *the* morphism $* \to *$ in 1, and let Y be an object in $\mathbb{B}$ (vacuously above $*$). Then $\mathrm{id}_Y : Y \to Y$ is a Cartesian lift of $* \to *$, whence $\mathbb{B} \to 1$ is a fibration.

Finally, let X and Y be sets and $p : X \to I$ a function, considered as a functor between discrete categories. All morphisms in I and X are identities, so if $i \in I$ and $x \in X$ is a "lift" of i (i.e., $p(x) = i$ or equivalently $x \in p^{-1}(i)$), then id_x is a Cartesian lift of id_i. Therefore, $p : X \to I$ is a fibration. $\square$

1.1.8. For an arbitrary category $\mathbb{B}$, consider the domain functor $\mathrm{dom} : \mathbb{B}^{\to} \to \mathbb{B}$.

(i) Describe the fibre category above $I \in \mathbb{B}$. It is usually called the **opslice category** or simply **opslice** and written as $I \setminus \mathbb{B}$.

(ii) Show that dom is a fibration (without any assumptions about $\mathbb{B}$).

(iii) Show also that for each $I \in \mathbb{B}$ the domain functor $\mathrm{dom}_I : \mathbb{B}/I \to \mathbb{B}$ is a fibration.

Solution, (i). For a category $\mathbb{B}$ and an object $I \in \mathbb{B}$, the category $I \setminus \mathbb{B}$ has

objects Morphisms $\begin{pmatrix} I \\ \downarrow\varphi \\ X \end{pmatrix}$ in $\mathbb{B}$ with domain I.

morphisms Given objects $\begin{pmatrix} I \\ \downarrow\varphi \\ X \end{pmatrix}$ and $\begin{pmatrix} I \\ \downarrow\psi \\ Y \end{pmatrix}$ in $I \setminus \mathbb{B}$, a morphism in $I \setminus \mathbb{B}$ from the first to the second consists of a morphism $f : X \to Y$ in $\mathbb{B}$

such that the following diagram commutes.

$$\begin{tikzcd} & I \arrow[dl, "\varphi"'] \arrow[dr, "\psi"] & \\ X \arrow[rr, "f"] & & Y \end{tikzcd}$$

□

Solution, (ii). Consider a morphism $u : I \to J$ in $\mathbb{B}$, and another morphism $\psi : J \to Y$ in $\mathbb{B}$ (i.e., an object $\begin{pmatrix} J \\ \downarrow\psi \\ Y \end{pmatrix}$ in $\mathbb{B}^{\to}$ over J with respect to $\mathrm{dom} : \mathbb{B}^{\to} \to \mathbb{B}$). We claim that

$$\begin{pmatrix} I \\ \downarrow \psi \circ u \\ Y \end{pmatrix} \xrightarrow{(\mathrm{id}_Y, u)} \begin{pmatrix} J \\ \downarrow \psi \\ Y \end{pmatrix} \qquad \text{i.e.,} \qquad \begin{tikzcd} I \arrow[r, "u"] \arrow[d, "\psi\circ u"'] & J \arrow[d, "\psi"] \\ Y \arrow[r, equal] & Y \end{tikzcd} \tag{1.1.8.1}$$

is a Cartesian lift of u. Indeed, consider a situation as in the following commutative diagram of solid arrows.

$$\begin{tikzcd} \begin{pmatrix} K \\ \downarrow \varphi \\ Z \end{pmatrix} \arrow[rr, bend left, "(g, u\circ v)"] \arrow[dr, dashed, "?"] \arrow[dd, "\mathrm{dom}"'] & & \\ & \begin{pmatrix} I \\ \downarrow \psi \circ u \\ Y \end{pmatrix} \arrow[r, "(\mathrm{id}_Y, u)"] \arrow[dd, "\mathrm{dom}"] & \begin{pmatrix} J \\ \downarrow \psi \\ Y \end{pmatrix} \arrow[dd, "\mathrm{dom}"] \\ K \arrow[dr, "v"'] & & \\ & I \arrow[r, "u"] & J \end{tikzcd} \tag{1.1.8.2}$$

We must find a unique morphism in $\mathbb{B}^{\to}$ to fill in the dashed "?" arrow in (1.1.8.2) that makes the diagram commute. But observe that (1.1.8.2) is

equivalent to the following diagram.

$$\begin{tikzcd}
K \arrow[r, "v"] \arrow[d, "\varphi"] & I \arrow[r, "u"] \arrow[d, "\psi\circ u"] & J \arrow[d, "\psi"] \\
Z \arrow[r, dashed, "?"] \arrow[rr, bend right, "g"'] & Y \arrow[r, equal] & Y
\end{tikzcd}$$

It becomes evident that the unique morphism that fills in the dashed "?" arrow in (1.1.8.2) to makes the diagram commute is $? = (g, v)$. Thus, (1.1.8.1) is a Cartesian lift of u, and hence $\mathrm{dom} : \mathbb{B}^{\to} \to \mathbb{B}$ is a fibration. $\square$

Solution, (iii). The same proof as in (ii) applies here as well. Explicitly, fix an object I in $\mathbb{B}$, let $u : X \to Y$ be a morphism in $\mathbb{B}$, and let $\begin{pmatrix} Y \\ \downarrow\psi \\ I \end{pmatrix}$ in $\mathbb{B}/I$ over Y relative to the functor $\mathrm{dom}_I : \mathbb{B}/I \to \mathbb{B}$. Then a straightforward modification of the proof of (ii) shows that

$$\begin{pmatrix} X \\ \downarrow\psi\circ u \\ I \end{pmatrix} \xrightarrow{u} \begin{pmatrix} Y \\ \downarrow\psi \\ I \end{pmatrix} \qquad \text{i.e.,} \qquad \begin{tikzcd}
X \arrow[rr, "u"] \arrow[dr, "\psi\circ u"'] & & J \arrow[dl, "\psi"] \\
& I &
\end{tikzcd}$$

is a Cartesian lift of u, and hence $\mathrm{dom}_I : \mathbb{B}/I \to \mathbb{B}$ is a fibration. $\square$

1.1.9. Assume $\mathbb{B}$ is a category with pullbacks. Show that the functor $\mathbb{B}^{\to\to} \to \mathbb{B}^{\to}$ sending $\xrightarrow{f}\xrightarrow{g}$ to $\xrightarrow{g}$ is a fibration. Is the composite $B^{\to\to} \to \mathbb{B}^{\to} \to \mathbb{B}$ also a fibration?

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